1 Transpose of a Linear Map

Definition 1.1 (Transpose of a Linear Map)**.** Let V and W be vector spaces over F, and let $T \in \mathscr{L}(V,W)$. The *transpose of* T (or *dual map of* T) is the linear map $T^t \in \mathscr{L}(W^*,V^*)$ define for each $f \in W^*$ by

$$T^*(f) = f \circ T.$$

That is, for each linear functional $f \in W^*$, $T^*(f)$ is the linear functional in V^* defined by $T^*(f)(v) = f(Tv)$ for each $v \in V$.

Exercise 1.2. Since $f \in W^* = \mathscr{L}(W,F)$ and $T \in \mathscr{L}(V,W)$, because the composition of linear maps is linear we indeed have $f \circ T \in \mathscr{L}(V,F) = V^*$ for each $f \in W^*$, which shows that T^t indeed maps W^* to V^*. Show that $T^t : W^* \to V^*$ is linear.

Example 1.3. Consider the differentiation map $D : \mathscr{P}(\mathbb{R}) \to \mathscr{P}(\mathbb{R})$ defined by $Dp = p'$. Fix $t \in \mathbb{R}$ and let $\mathrm{ev}_t \in \mathscr{P}(\mathbb{R})^*$ be the evaluation map which sends $p \mapsto \mathrm{ev}_t(p) = p(t)$. Then

$$D^t(\mathrm{ev}_t)(p) = \mathrm{ev}_t(Dp) = p'(t).$$

Thus, $D^t(\mathrm{ev}_t) \in \mathscr{P}(\mathbb{R})^*$ is the linear functional taking $p \mapsto p(t)$.

Exercise 1.4. Let D again be the differentiation operator on $\mathscr{P}(\mathbb{R})$. Let $\varphi \in \mathscr{P}(\mathbb{R})^*$ be the definite integral $\varphi(p) = \int_0^1 p(t)dt$. What is $D^t(\varphi)$?

Exercise 1.5. Let $\mathrm{id}_V \in \mathscr{L}(V)$ be the identity map on V. What is $\mathrm{id}_V^t \in \mathscr{L}(V^*)$? Similarly, if 0_V is the zero map on V, what is 0_V^t?

Proposition 1.6 (Properties of the transpose)**.** *For all linear maps T_1, T_2, T and $\alpha \in F$,*

(a) $(T_1 + T_2)^t = \alpha T_1^t + T_2^t$ *(that is, the map $T \mapsto T^t$ is linear)*

(b) $(T_1 T_2)^t = T_2^t T_1^t$ *(where $T_1 T_2$ denotes composition)*

Exercise 1.7. Prove Proposition 1.6.

2 Matrix of the Transpose of a Linear Map

Let V, W be finite-dimensional vector spaces. Choose ordered bases $B_V = (v_1, \ldots, v_n)$ and $B_W = (w_1, \ldots, w_m)$ of V and W. Let $B_V^* = (f_1, \ldots, f_n)$ and $B_W^* = (g_1, \ldots, g_m)$ be ordered bases of V^* and W^*, respectively.

Proposition 2.1. *The matrix of T^t is the transpose of the matrix T. That is,*

$$[T^t]_{B_W^*, B_V^*} = ([T]_{B_V, B_W})^t.$$

Proof. As for any linear map, the jth column of $T^t : W^* \to V^*$ is the coordinate vector with respect to B_V^* of $T^t(g_j)$:

$$T^t(g_j)(v_i) = (\sum_{k=1}^{m} [T^t]_{kj} f_k)(v_i) = \sum_{k=1}^{m} [T^t]_{kj} f_k(v_i) = \sum_{k=1}^{m} [T^t]_{kj} \delta_{ki} = [T^t]_{ij}.$$

On the other hand, by definition of T^t we have

$$T^t(g_j)(v_i) = g_j(Tv_i) = g_j\left(\sum_{k=1}^{m} [T]_{ki} w_k\right) = \sum_{k=1}^{m} [T]_{ki} g_j(w_k) = \sum_{k=1}^{m} [T]_{ki} \delta_{jk} = [T]_{ji} = ([T]^t)_{ij}.$$

Thus, $[T^t] = [T]^t$. □

Exercise 2.2. Let V and W be finite-dimensional vector spaces, $T \in \mathscr{L}(V, W)$, and $g \in W^*$. Choose ordered bases $B_V = (v_1, \ldots, v_w)$ and $B_W = (w_1, \ldots, w_m)$ for V and W, respectively, and let $B_V^* = (f_1, \ldots, f_n)$ and $B_W^* = (g_1, \ldots, g_m)$ be the corresponding dual bases of V^* and W^*. Prove that $[T^t g]_{B_V^*, S} = [g]_{B_W^*, S}[T]_{V,W}$, where S denotes the standard ordered basis (1) of F.

Remark 2.3. One can write the result of Exercise 2.2 in terms of column vectors rather than row vectors by taking the transpose of both sides of $[T^t g] = [g][T]$, which gives $[T^t g]^t = [T]^t [g]^t = [T^t][g]^t$.

Example 2.4. We showed before that the matrix of the differentiation map $D : \mathscr{P}_3(\mathbb{R}) \to \mathscr{P}_2(\mathbb{R})$ defined by $Dp = p'$ is given, with respect to the standard bases of these spaces,

$$[D] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

By Proposition 2.1, the matrix of D^t with respect to the dual bases is

$$[D^t] = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Fix $t \in \mathbb{R}$ and let $\mathrm{ev}_t \in \mathscr{P}_3(\mathbb{R})^*$ be the evaluation map defined by $\mathrm{ev}_t(p) = p(t)$. The dual basis $B^* = (f_0, f_1, f_2)$ of $B = (1, x, x^2)$ is defined by $f_i(x^j) = \delta_{ij}$. The matrix of ev_t with respect to this basis is $[\mathrm{ev}_t] = \begin{bmatrix} \mathrm{ev}_t(1) & \mathrm{ev}_t(x) & \mathrm{ev}_t(x^2) \end{bmatrix} = \begin{bmatrix} 1 & t & t^2 \end{bmatrix}$. The matrix of $D^t(\mathrm{ev}_t) \in \mathscr{P}_3^*(\mathbb{R})$ is therefore given by

$$[D^t(\mathrm{ev}_t)] = [\mathrm{ev}_t][D] = \begin{bmatrix} 1 & t & t^2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2t & 3t^2 \end{bmatrix}.$$

If $p = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \in \mathscr{P}_3(\mathbb{R})$, then we have

$$[D^t(\mathrm{ev}_t)][p] = \begin{bmatrix} 0 & 1 & 2t & 3t^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = [a_1 + 2a_2 t + 3a_3 t^2] = [p'(t)].$$

Exercise 2.5. Find the matrix of $[D^t(\varphi)]$, where $\varphi \in \mathscr{P}_3(\mathbb{R})^*$ is defined as $\varphi(p) = \int_0^1 p(t)dt$ as in Exercise 1.4. Show that $[D^t\varphi][p]$ gives the expected result for $p \in \mathscr{P}_3(\mathbb{R})$.

3 Kernel and Image of the Transpose of a Linear Map

Let V and W be finite-dimensional vector spaces and $T \in \mathscr{L}(V,W)$. Then $T^t \in \mathscr{L}(W^*,V^*)$ and, by choosing bases of V and W and the corresponding dual basis of V^* and W^*, we have with respect to these bases

$$[T^t] = [T]^t.$$

We have seen before that the rank of a matrix and its transpose are equal, so

$$\text{rank}T^t = \text{rank}T \leq \min\{m,n\}.$$

Exercise 3.1. Prove that

(a) T is injective if and only if T^t is surjective.

(b) T is surjective if and only if T^t is injective.